EGIN

A cross-platform mobile and web application for managing a land plot

Probation Period 1 · Concise Report

Group: SE-2414

Team: Adilzhan Amangeldi · Rustam Seitzhan

Representative: Adilzhan Amangeldi

Supervisor: Shuteyeva Gulnur Serikovna

Project version: prob-period-1 · 614ca67
11 sections · key diagrams · prototype · verification and next steps

23 October 2026 · SE-2414

Contents

1. Project Proposal 3
2. Evidence of the Relevance 4
3. Literature Review 5
4. Existing System Comparison 6
5. Technology Comparison 7
6. Requirements Specification 8
7. Architecture and Diagrams 9
8. UX and Prototype 12
9. Verification and Evaluation Plan 13
10. Project Management and Risks 14
11. References 15

Prototype and supporting materials

Prototype · Developer portal · Frozen project source · Editable summary diagrams · Full architecture diagrams · Screenshots and capture records · Verification records · Editable concise report

This concise report describes the frozen `prob-period-1` implementation. Detailed specifications and historical evidence remain linked; live websites may change. Implementation, executed checks and planned acceptance are distinguished throughout. References is section 11, following the source-document order.

1. Project Proposal

Detailed section and supporting evidence

EGIN is a cross-platform web and mobile application for managing agricultural land plots in Kazakhstan. Group **SE-2414** comprises **Adilzhan Amangeldi** and **Rustam Seitzhan**, both Full-Stack Developers. The supervisor is **Shuteyeva Gulnur Serikovna**.

Small farms obtain boundaries, weather and accounting information from separate systems, while unreliable rural connectivity restricts cloud-only workflows [1–3]. Primary users are farm owners, managers, agronomists and field workers; secondary users include accountants and integration developers.

Aim. Bring plot boundaries, coordinate-based forecasts, available field information, sensor registration and reports into one interface, retaining supported local data offline and synchronising changes safely.

The implementation includes cadastral lookup through Kazakhstan's public EGKN map, GeoJSON import/export, IndexedDB storage, explicit synchronisation conflicts, passkey authentication and recovery, cross-device QR sign-in, read-only 1C OData import and a scoped external API. The web/PWA version is available; Capacitor configuration supports the intended mobile delivery path. A completed native release is not claimed.

Adilzhan's documented responsibilities cover cadastral GIS, the crop scene and weather, 1C, and the developer API/portal. Rustam's cover synchronisation, authentication, security, CI/CD and blue/green deployment.

Current boundaries. Sensor telemetry, validated crop-growth analytics, eGov authentication and contractual cadastral integration remain pending or excluded. The 3D crop scene is illustrative. The event centre preserves existing records but has no active manual diary composer. Chat, marketplace and business extensions exist beyond the core diploma scope.

The next objectives are to substantiate user needs, close outstanding verification evidence and prepare tested mobile delivery. Expected outputs are the application, traceable requirements, editable architectural diagrams and a reproducible evaluation record.

2. Evidence of the Relevance

Detailed section and supporting evidence

The team reports informal discussions with approximately **10–15 farmers and agronomists in Kazakhstan**, including people managing substantial agricultural holdings. These discussions concerned the project idea and agricultural problems. Individual dates, transcripts and attributable findings are not attached, so this estimate must not be presented as completed structured interviews.

The team also reports meeting **Zhiguli Dairabayev** and receiving support for promoting EGIN. His official parliamentary profile [19] establishes background, not evidence of this particular meeting. No formal partnership, signed client agreement or software acceptance is claimed.

The evidence therefore has three distinct strengths: literature supports the broader problem [1–3]; the team reports stakeholder contact; the repository and prototype demonstrate software that can be evaluated. None independently proves adoption or improved farm outcomes.

Five hypotheses guide follow-up work: saved information remains useful during connection loss; cadastral lookup reduces boundary-entry effort; passkey recovery is understandable; compatible 1C import reduces duplicate entry; and plot-specific forecasts simplify information retrieval. These are proposed hypotheses, not reconstructed interview findings.

Next evidence required: document 3–5 substantive discussions with dates, participant context, permission and requirement links; attach meeting-specific material where available; and run paired tasks measuring completion, errors and time. Small-farm participants should be distinguished from larger-holding managers. Current status is **engagement reported; supporting records and feature-level validation pending**.

3. Literature Review

Detailed section and supporting evidence

The detailed review examines **13 sources** across agricultural adoption, Kazakhstan's context, sensors and digital twins, local-first systems, authentication and weather. It combines recent publications with foundational synchronisation research, an FAO report and the weather provider's software reference. The evidence informs design choices; it does not establish EGIN's effectiveness with users.

Research theme and sources	Main finding	Implication and limitation for EGIN
Adoption and Kazakhstan [1–3]	Connectivity, cost and skills constrain uptake.	Prioritise offline access and simple onboarding; sector findings require local validation.
IoT and digital twins [4–7]	Useful models require reliable measurements and integration.	Keep sensor registration separate from telemetry; the crop illustration is not a validated twin.
Local-first and CRDTs [8–9]	Local copies support availability; convergence and merge design require trade-offs.	Use IndexedDB, versions, tombstones and explicit conflict choices; validate recovery behaviour.
Passkeys [10–11]	Recovery, cross-device friction and key lifecycle failures matter.	Rotate recovery codes and protect the last passkey; security mechanisms still need usability evaluation.
Rural PWA assistance [12]	Lightweight web delivery can suit constrained environments.	Supports PWA exploration; an advisory chatbot is not direct evidence for cadastral workflows.
Open-Meteo [13]	Shared forecast models supply coordinate-based weather.	Show source and freshness; forecast output is not an on-site measurement.

Three conclusions shape the implementation. First, the offline-first approach responds to documented access barriers, while its value must be measured on actual farm tasks. Second, explicit data provenance prevents an attractive visualisation from implying unavailable agricultural analytics. Third, authentication and synchronisation require failure-oriented verification as well as successful demonstrations.

The review's principal limitation is transferability: general agricultural surveys, security studies and distributed-systems theory do not replace participant observations, physical-device checks or a controlled evaluation of this particular application.

4. Existing System Comparison

Detailed section and supporting evidence

The source review compares **EGIN v0.3** [18] with **Cropwise Operations** [14], **EOSDA Crop Monitoring** [15], **AGRIVI** [16] and **AgroSignal.Expert** [17]. The compact matrix retains four common criteria from the detailed comparison. Competitor entries summarise the reviewed product documentation, not hands-on comparative testing.

Criterion	EGIN	Cropwise	EOSDA	AGRIVI	AgroSignal
Offline support	Saved local data	Mobile scouting	Mobile scouting	Mobile records	Mobile scouting
Plot weather	Coordinate forecast	Field forecast	Field forecast	Field forecast	Weather module
Satellite indices	Outside scope	Available	Available	Available	Available
Built-in KZ EGKN lookup	Implemented	Not documented	Not documented	Not documented	Not documented

EGIN’s proposed niche is convenient access to Kazakhstan’s cadastral boundaries alongside local storage, forecasts and integration with existing accounting data. Competitor boundary workflows described in the source comparison use drawing or imported geographic files. “Not documented” is a limit of the reviewed evidence, not proof that a capability could never exist.

Established products offer broader satellite monitoring and commercial integrations. EGIN currently provides a sensor registry rather than live telemetry, and its illustrative 3D scene does not replace those capabilities.

EGIN’s browser entry lowers the installation barrier. Capacitor mobile delivery remains a release-stage objective. Offline, security and integration mechanisms exist in the source, but comparative speed, usability, agricultural benefit and complete end-to-end acceptance have not yet been measured.

5. Technology Comparison

Detailed section and supporting evidence

Six decisions were compared against offline use, implementation reuse, team capacity and operational complexity. The stack supports the website and future mobile packaging; Capacitor configuration does not establish accepted Android/iOS builds.

Decision	Selected approach and rationale	Main alternative or limitation
Client platform	React/TypeScript, PWA and Capacitor reuse the interface, mapping and Three.js scene.	Kotlin/Swift clients increase maintenance; Flutter/React Native requires adapting the web core.
Core server/store	Node.js/Express with SQLite WAL simplifies single-host transactions and backup.	SQLite has a single-writer ceiling. PostgreSQL supports additional modules outside the core diploma datastore.
Offline synchronisation	IndexedDB, sequence cursors, versions, tombstones and explicit conflict resolution make retries and conflicts inspectable.	Last-write-wins risks silent overwrites; CRDTs introduce complexity for structured records. Data-preservation acceptance still requires testing.
Authentication	Passkeys, rotating recovery codes and separate-device QR login avoid password handling.	Native iOS configuration remains pending; official eGov integration is unavailable.
Plot mapping	Leaflet and the public EGKN adapter provide cadastral lookup; GeoJSON and geolocation provide fallbacks.	The upstream public interface may change; contractual LandCoordinates access remains pending.
Weather	Open-Meteo supplies coordinate-based forecasts without adding a separate sensor system.	Forecasts are model output, not field measurements; commercial-use terms must be checked before commercial release.

These are architectural trade-offs, not measured product superiority. The detailed comparison records alternatives; Sections 6 and 9 specify acceptance and verification.

6. Requirements Specification

Detailed section and supporting evidence

Requirements retain stable `FR` and `NFR` identifiers and MoSCoW priorities. The complete register contains **39 Must requirements: 21 functional and 18 non-functional**. This is a priority count, not a pass count. Implementation status comes from source inspection; acceptance additionally requires the specified automated test, demonstration, inspection or measurement.

The grouped register below summarises key acceptance obligations; the full specification retains individual priorities, criteria and traceability to literature, stakeholder hypotheses and recorded owner constraints.

Requirement group and IDs	Acceptance criteria	Current evidence/status
Identity: FR-AUTH-1–5	Reject invalid passkey assertions; rotate recovery codes; prevent last-key deletion; distinguish EGIN QR from unavailable eGov.	Web implementation present; native iOS passkey configuration pending. Full authentication acceptance remains open.
Plots: FR-FLD-1,3,4,6	Preview before saving; exclude personal cadastral data; preserve GeoJSON geometry; retain boundaries offline.	Implemented. Geometry/privacy fixtures passed; complete browser/ device demonstration remains pending.
Offline data: FR-SYNC-1–8; FR-REC-1	Preserve queued changes; resolve conflicts explicitly; avoid duplicates and resurrected deletions; isolate profiles.	Synchronisation mechanics implemented; manual diary creation is partial because no composer is exposed.
Weather, events and exports: FR-WTH-1–3; FR-EVT-1; FR-RPT-1	Bind forecasts to plot coordinates, filter events correctly and export safe HTML/ CSV/JSON.	Implemented; relevant unit checks passed. Live-provider and export-download acceptance remain separate.
Honest presentation: FR-HOME-3; FR-SNS-2	Label the crop scene as illustrative; never imply telemetry from sensor registration.	Implemented; fixture screens show these boundaries.
Integrations: FR-INT-1–2; FR-API-1–3	Keep 1C import read-only/idempotent; enforce scoped keys and route-catalog consistency.	Implemented artefacts; full integration suites remain blocked or pending.
Quality: NFR-USE-1–3; NFR-PERF-1; NFR-REL-1–3	No phone overflow; readable contrast; LCP ≤4 s; no silent data loss; rollback ≤60 s.	Comprehensive accessibility, performance, recovery and device measurements planned.
Security and delivery: NFR-SEC-2,3,6; NFR-CMP-1; NFR-MNT-1,3	Enforce ownership, protect secrets, separate environments, protect merges and gate native builds.	Safeguards inspected; branch protection was absent at historical review; native acceptance remains open.

Out-of-scope requirements include sensor telemetry and contractual government/partner integrations. A completed web implementation does not close partial Must requirements. Outstanding criteria need recorded results or an explicitly agreed scope revision; Section 9 distinguishes executed checks from planned acceptance.

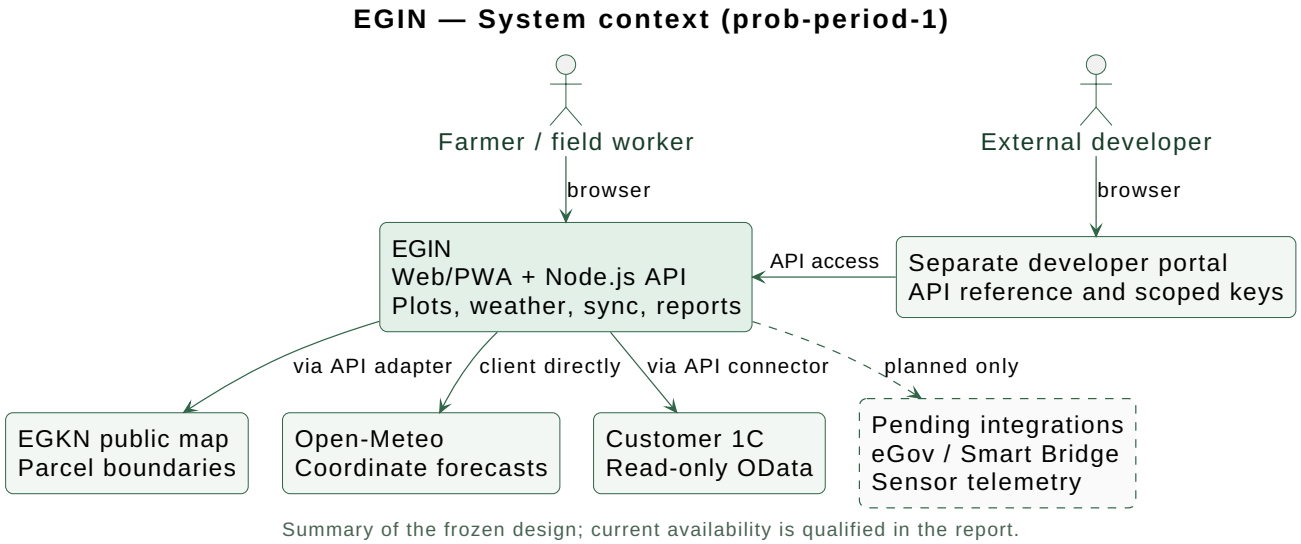
7. Architecture and Diagrams

Detailed section and supporting evidence

EGIN uses a local-first client–server architecture. The React application keeps the user's working data in IndexedDB; the Node.js API provides identity, synchronisation and integration services. SQLite stores core account and plot data. A separate developer portal exposes documented, scoped access. Additional chat/trade modules are outside this core diagram set.

7.1 System context

The farmer works with plots, weather and local records. The cadastral adapter obtains public parcel data; 1C supplies read-only accounting imports. Weather is requested directly by the client. External developers use the documented API. eGov and live sensor telemetry remain pending integrations.

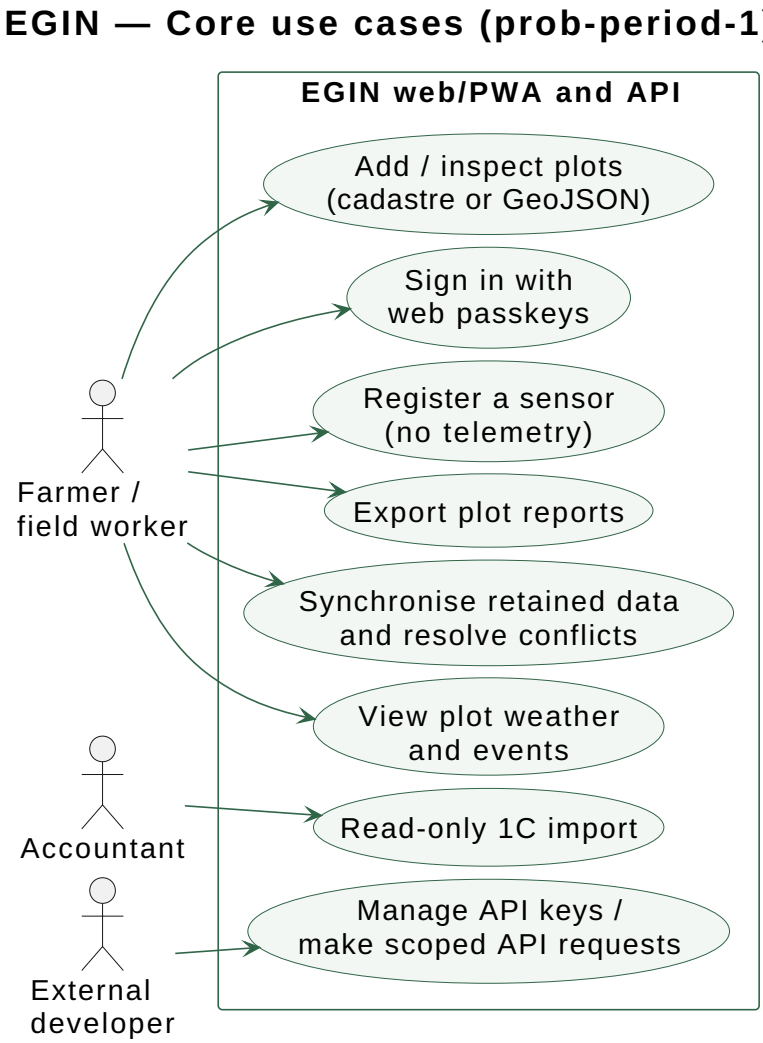


System context: users, EGIN and external services. Editable PlantUML source

Summary diagram. The complete system context and integration contracts remain in the detailed architecture.

7.2 Core use cases

The diagram groups the main user goals. Plot acquisition and forecast viewing are the principal walkthrough. Authentication controls server access; local guest data is separate from authenticated synchronisation. The current event centre can display retained history, but the legacy manual diary composer is not exposed in the active UI.

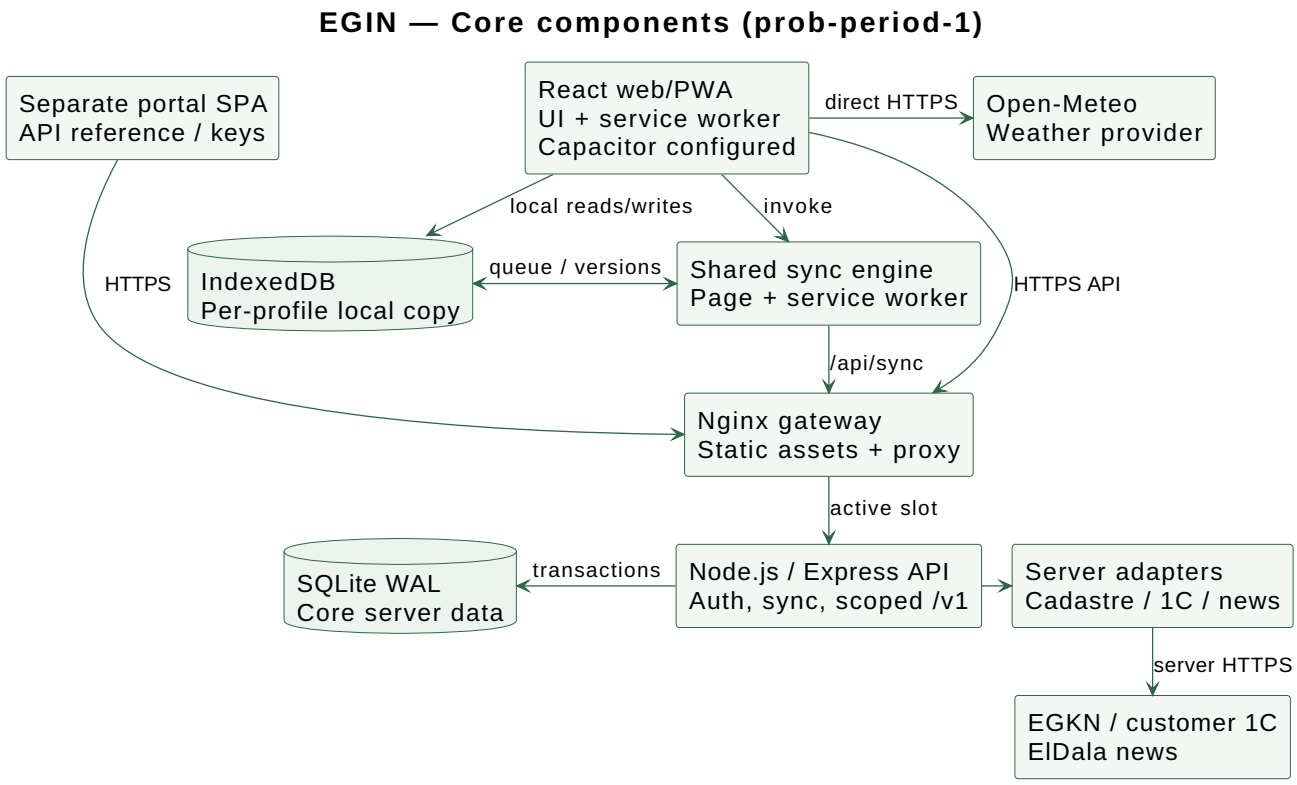


Core use cases and their actors. Editable PlantUML source

User goal	Requirement trace
Sign in, recover access and synchronise	FR-AUTH-1–4; FR-SYNC-2–8
Acquire a plot and view its forecast	FR-FLD-1–7; FR-WTH-1–3
Review events, register a sensor and export	FR-EVT-1; FR-SNS-1–2; FR-RPT-1
Import 1C records or integrate through the API	FR-INT-1–2; FR-API-1–3

7.3 Components and data flow

The client and service worker share one synchronisation engine. Changes use monotonic sequence cursors, record versions, deletion tombstones and explicit conflict resolution. This is designed to preserve concurrent work; successful end-to-end acceptance remains a separate check.



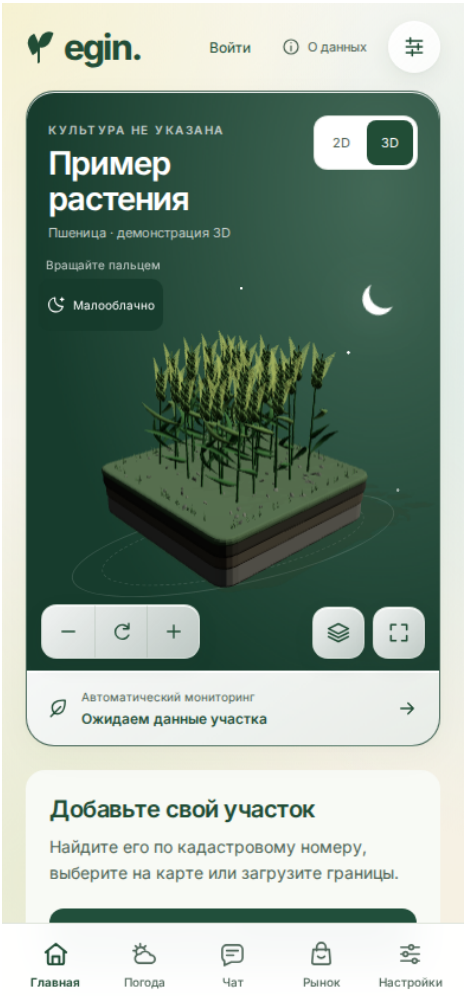
Core components and data stores. Editable PlantUML source

The Nginx gateway serves both web applications and routes API traffic to the active deployment slot. Main-app and portal sessions remain separate. Capacitor configuration shares the web codebase, while native release validation is pending. Provider adapters apply bounded requests and privacy filtering. The full component, deployment and sequence diagrams retain implementation-level detail and editable PlantUML sources.

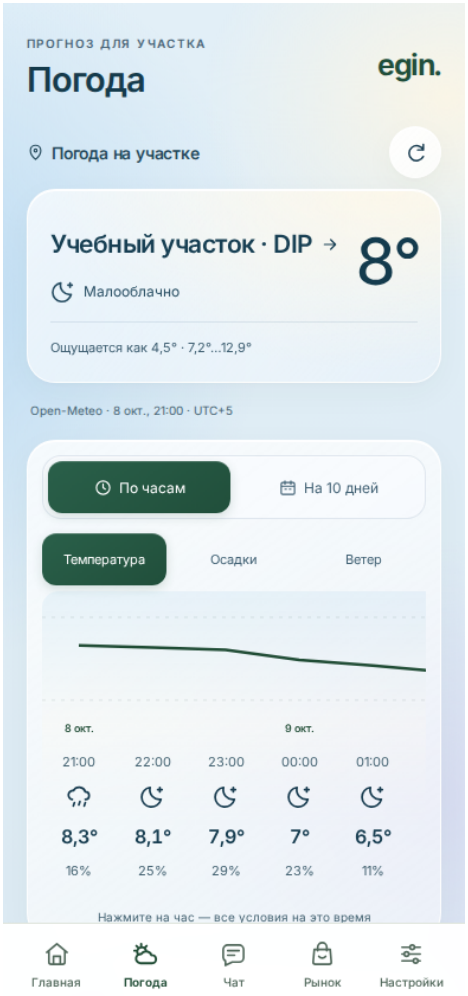
8. UX and Prototype

Detailed section and supporting evidence

The web prototype uses a phone-first layout with Home, Weather, Chat, Market and Settings tabs. Plots, events, reports and the sensor registry are reached from Home or Settings. The interface is Russian; report descriptions are English. The crop illustration is an interface example, not a measured digital twin.



Home: crop illustration and plot onboarding.



Weather: the selected plot and fixture forecast.

The core demonstration is: **open Home → add or import a plot → confirm its boundary → inspect the forecast → review events → export available data**. A separate authenticated exercise must check offline changes, reconnection and conflict resolution. The developer portal provides API examples and scoped-key management.

These images come from fresh web builds of `b0b0f46`, whose application tree equals the tagged version. Nine captures completed on 23 October 2026 with zero page errors. Synthetic guest records and intercepted APIs were used; the forecast is a historical fixture. These captures do not establish live provider access, authentication, offline reload, export download or user acceptance. All screenshots, source fingerprints and reproduction instructions are available separately.

9. Verification and Evaluation Plan

Detailed section and supporting evidence

Verification checks requirements; evaluation tests whether users complete useful tasks more effectively. Evidence below separates historical execution at `8cc0ca3` from subsequent web builds at `b0b0f46`. Their application-source trees are identical. Neither screenshots nor an implemented test file demonstrate complete acceptance.

Check	Recorded result and interpretation
Historical TypeScript/unit checks	Both web apps passed typechecking. Mobile unit results: **73 passed, 0 failed, 0 skipped**. Coverage includes geometry, weather, events, exports and synchronisation helpers.
Historical API run	**38 passed, 20 failed, 1 skipped**; overall command failed. All 20 failures arose during `sharp` native-runtime import, before affected assertions executed. Counts combine tests and failed file-level entries; they are not requirement coverage.
Historical CI snapshot	Mobile checks succeeded, with Android packaging skipped. Chat and Trade/billing/video workflows failed; Market was cancelled during browser installation. These outcomes remain unresolved in this evidence set.
Subsequent fresh web builds	Mobile and developer TypeScript/Vite builds passed; mobile service-worker generation passed. Missing build dependencies were supplied in an isolated copy. This did not repair or rerun the blocked API suite.
Subsequent UI capture	**9 screenshots, 0 reported page errors**, Chromium 153.0.8010.12, 23 October 2026. Synthetic guest records and intercepted APIs establish rendering, not live authentication, synchronisation, downloads or service-worker behaviour.

Evidence is retained in the local verification record, fresh-build record, capture metadata and detailed evaluation/CI register. Database/inference-dependent scenarios and physical-device certification are not established by these runs.

Planned outcome evaluation — all three targets remain unmeasured:

1. **Task completion:** at least 90% correct, unassisted completions across the defined task set; report counts and every failure.
2. **Parcel-addition time:** median ≤60 seconds and lower than a matched manual workflow; retain failures and timeouts alongside successful timings.
3. **Offline preservation:** 100% of queued operations accounted for, with zero silent losses or duplicates across at least 20 interruption/reconnection cycles.

Freeze the candidate revision, devices, fixtures and network conditions. Compare equivalent tasks using separate cadastral/weather tools and a record template; retain consented observations and counterbalance task order. First rerun blocked API/browser checks, then complete physical-device, accessibility and recovery checks. Native release readiness, yield improvements, adoption and user satisfaction are not claimed.

10. Project Management and Risks

Detailed section and supporting evidence

The repository contains the implemented web core, deployment tooling and additional chat, marketplace and business modules. Fresh web builds and nine fixture screenshots are recorded. User-research documentation, complete acceptance evidence and native release remain unfinished. Historical API/CI failures are retained as failures; successful rendering does not close those work items.

Documented responsibilities: Rustam Seitzhan owns synchronisation, authentication, security and deployment; Adilzhan Amangeldi owns cadastral GIS, weather/3D, 1C and the developer API. Research coordination, final evaluation and extension ownership still need agreement. The supervisor is Shuteyeva Gulnur Serikovna.

GitHub history and the report's local work register provide management evidence. No actual YouTrack board is evidenced. Local `PM-` identifiers are proposed references, not historical tracker issues. No approved calendar deadlines were supplied. Remaining tasks should receive a named owner, priority, deadline, dependency and acceptance evidence before being marked Done.

Completed implementation example — PM-02: cadastral parcel passports. Commit `79468ce` adds public EGKN details and offline snapshots. Adilzhan is the documented responsible developer. Expected outcome: users can inspect a saved parcel's public information with its source and capture time. Proposed acceptance, derived from FR-FLD-3/6/7: exclude personal identifiers; explicitly mark missing data; retain saved contours offline; label registered and calculated areas separately. Implementation is present; full device/offline acceptance remains separate evidence. Task deadline and original priority were not recorded.

Risk / current limitation	Required next action
User discussions lack traceable records	Collect 3–5 consented records and link findings to requirements.
Failed checks or lost/conflicting data	Repair isolated test dependencies; rerun security, synchronisation and affected CI checks.
Public data providers change or fail	Verify bounded failures, GeoJSON fallbacks and clearly labelled cached data.
Scope expands beyond evaluated core	Confirm core acceptance scope and assign extension ownership separately.
Device/native readiness is overstated	Run physical-device checks; prepare Capacitor release only after required configuration and validation.

The next stage proceeds through evidence completion, verification, user/device evaluation and release readiness. Each decision should retain the chain **requirement → work item → commit → check result → acceptance**, with unresolved conditions visible.

11. References

Detailed section and supporting evidence

Numbering follows the detailed source bibliography. Product-documentation access dates are 22 October 2026; the stakeholder profile was consulted on 23 October 2026.

Full publication metadata remains available in the detailed bibliography; abbreviated author lists below retain original source identities, titles and publication years.

[1] Gumbi N. et al. *Towards Sustainable Digital Agriculture for Smallholder Farmers: A Systematic Literature Review.* Sustainability, 2023, 15(16), 12530. DOI.

[2] Abdrazakova A., Majerová I., Abdildin N. *Digital Agriculture: Challenges and Prospects in Kazakhstan.* Silesian University in Opava, 2022, pp. 1–15. ISBN 978-80-7510-527-1.

[3] FAO. *Promotion and Exchange of Good Practices on the Use of Digital Technologies in Agriculture: Republic of Kazakhstan.* April 2022. Repository.

[4] Ndjuluwa L.N.P. et al. *Internet of Things for Crop Farming: A Review of Technologies and Applications.* Commodities, 2023, 2(4), 367–381. DOI.

[5] Peladarinos N. et al. *Enhancing Smart Agriculture by Implementing Digital Twins: A Comprehensive Review.* Sensors, 2023, 23(16), 7128. DOI.

[6] Escribà-Gelonch M. et al. *Digital Twins in Agriculture: Orchestration and Applications.* Journal of Agricultural and Food Chemistry, 2024, 72(19), 10737–10752. DOI.

[7] Melesse T.Y. *Digital twin-based applications in crop monitoring.* Heliyon, 2025, 11(2), e42137. DOI.

[8] Kleppmann M. et al. *Local-first software: you own your data, in spite of the cloud.* Onward! 2019, pp. 154–178. DOI.

[9] Shapiro M. et al. *Conflict-free Replicated Data Types.* SSS 2011, LNCS 6976, pp. 386–400. DOI.

[10] Lassak L. et al. *Why Aren't We Using Passkeys? Obstacles Companies Face Deploying FIDO2 Passwordless Authentication.* 33rd USENIX Security Symposium, 2024.

[11] Jannett L. et al. *The State of Passkeys: Studying the Adoption and Security of Passkeys on the Web.* 35th USENIX Security Symposium, 2026.

[12] Tantaroudas N. et al. *Retrieval-grounded multilingual LLM assistance for island smallholder farmers.* Open Research Europe, 2026, 6, 45. DOI.

[13] Zippenfenig P. *Open-Meteo.com Weather API* [software]. Zenodo, 2023. DOI.

[14] Syngenta Digital. *Cropwise Operations: Field Management & Mobile Scouting Documentation.* Product.

[15] EOS Data Analytics. *EOSDA Crop Monitoring: Key Functions and Mobile Scouting App Documentation.* Product.

[16] AGRIVI. *Farm Management Software: Product Specifications and Mobile Application Guide.* Product.

[17] InfoBiS LLC. *AgroSignal.Expert and AgroSignal.Scouting User Manual.* Product.

[18] Seitzhan R., Amangeldi A. *EGIN: Mobile PWA for Land Plot Management — Project Repository and Technical Documentation*, version 0.3, 2026. Repository.

[19] Parliament of Kazakhstan. *Zhiguli Dairabayev: Official Parliamentary Profile, Eighth Convocation* (Kazakh). Profile. Background only; not evidence of the reported meeting or partnership.